

THERMAL LOAD CALCULATION

Al Reem Tower — Dubai Marina

MEP Estimation Department
SABI Contracting LLC

Date: April 8, 2026
Reference: TLC/ART/2026/001

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1. PROJECT SUMMARY

Parameter	Value
Project Name	Al Reem Tower
Client	Al Futtaim Group
Location	Dubai Marina, Plot 12-B
Building Type	Grade-A Commercial Office
Total Floors	24 + 4 Basement Parking
Total Area	185,000 sqft
Typical Floor Height	3.6 m
Consultant	KEO International Consultants
HVAC System	VRF (Variable Refrigerant Flow)
Total Cooling Load	1,035.4 kW (294.4 TR)
FAHU Load	180.2 kW
Net AC Load	855.2 kW (243.2 TR)

2. THERMAL LOAD SUMMARY BY FLOOR

Floor	Zone	Load (kW)	System	Indoor Units
Basement 1	Parking	0	Exhaust Only	0
Basement 2	Parking	0	Exhaust Only	0
Ground Floor	Lobby & Reception	85.5	VRF	12
Mezzanine	Meeting Rooms	42.3	VRF	8
1st Floor	Open Office	125.8	VRF	18
2nd Floor	Open Office	125.8	VRF	18
3rd Floor	Open Office	125.8	VRF	18
4th Floor	Open Office	125.8	VRF	18
5th Floor	Open Office	125.8	VRF	18
6th Floor	Executive Suites	98.4	VRF	14
7th Floor	IT & Server Room	180.2	Precision AC	6
Roof	Plant Room	0	N/A	0
TOTAL		1,035.4		130

3. HVAC SYSTEM CALCULATION

Step 1: Total Calculated KW

1,035.4 kW

Sum of all floor loads from thermal load drawing

Step 2: FAHU KW

180.2 kW

Fresh Air Handling Unit load (Server Room + common areas)

Step 3: AC Unit KW

855.2 kW

Total KW - FAHU KW = 1,035.4 - 180.2

Step 4: System Type

VRF

Based on equipment schedule — Variable Refrigerant Flow system

Step 5: Tonnage

243.2 TR

AC Unit KW / 3.517 = 855.2 / 3.517

Step 6: Total Tonnage

294.4 TR

Total KW / 3.517 = 1,035.4 / 3.517 (including FAHU)

4. PRICING FORMULA

Service	Rate	Area	Amount
HVAC (VRF)	AED 17.57/sqft	185,000 sqft	AED 3,250,000
Electrical	AED 10.00/sqft	185,000 sqft	AED 1,850,000
Plumbing	AED 4.97/sqft	185,000 sqft	AED 920,000
Fire Fighting	AED 4.22/sqft	185,000 sqft	AED 780,000
TOTAL MEP ESTIMATE			AED 6,800,000

5. EQUIPMENT SCHEDULE

Tag	Description	Model	Qty	Location
ODU-01 to ODU-08	VRF Outdoor Unit 22HP	Daikin RXYQ22TATL	8	Roof
FCU-GF-01 to 12	Indoor Unit Ducted 2.2kW	Daikin FXDQ20	12	Ground Floor
FCU-MZ-01 to 08	Indoor Unit Ducted 2.2kW	Daikin FXDQ20	8	Mezzanine
FCU-1F to 5F	Indoor Unit Ducted 2.2kW	Daikin FXDQ20	90	Typical Floors
FCU-6F-01 to 14	Indoor Unit Decorative 1.5kW	Daikin FXKQ15	14	6th Floor
FCU-7F-01 to 06	Precision AC 10kW	Emerson Liebert	6	7th Floor
FAHU-01 to 04	Fresh Air Handling Unit 5000CFM	Daikin AHU	4	Roof
EF-01 to EF-08	Exhaust Fan 2000CFM	Kruger	8	Basements

6. YARDSTICK COMPARISON

Market benchmark for Grade-A Office (Dubai): AED 32 - 42 per sqft

Our estimate: AED 36.76 per sqft — WITHIN RANGE

Status: Approved for quotation preparation

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